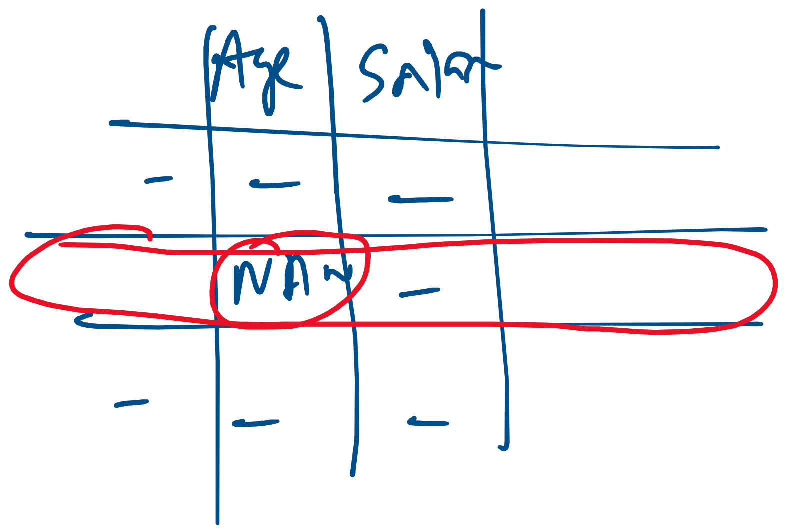
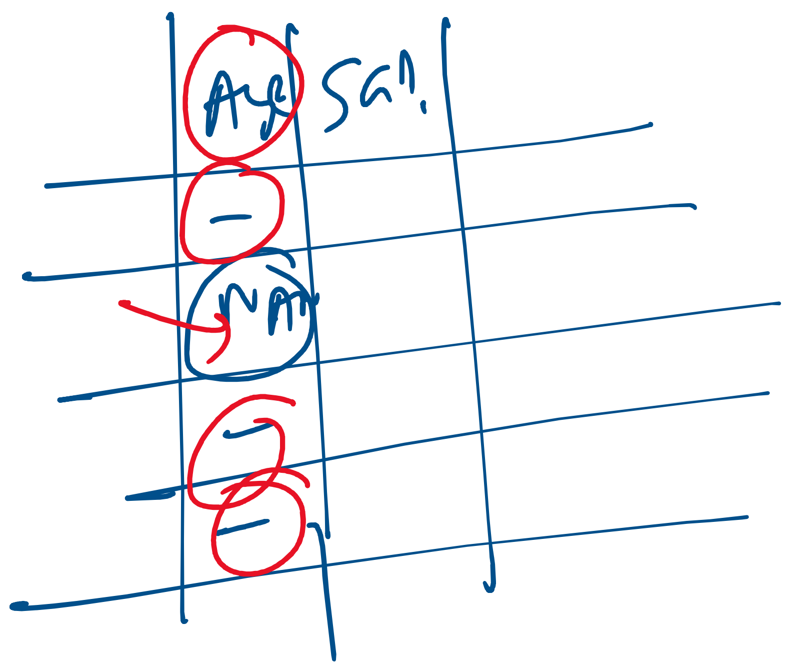
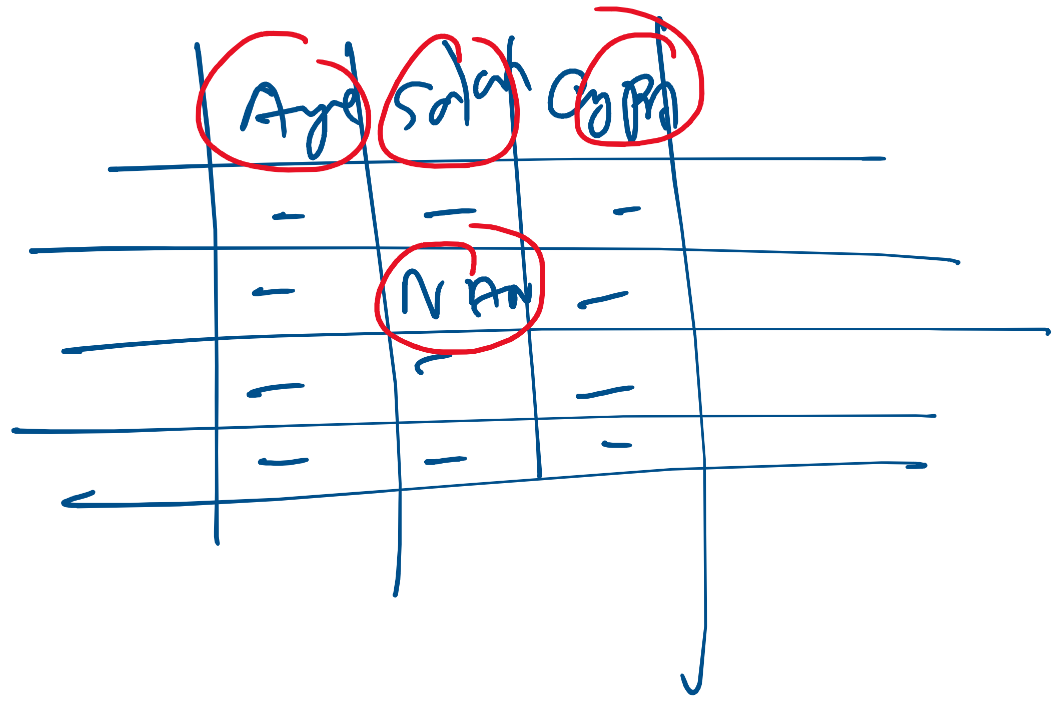
Handling missing data



Missing

- Remove them
- Impute (Fill)
  - Univariate
    - Fill
    - Focus one column
  - Multivariate
    - Fill
    - multi cols Fill up





multivariate

- → KNN Imputer
- → Iterative Im

Univariate

Numerical

→ mean/median

→ Arbitary (fill any value)

→ Random value

categorical → mode

→ Missing

Age
10
20
30
NAN
40
50

Remove them → CCA → complete case analysis

↓

discard → row → if colum values missing

CCA → work → row → missing value (NOT)

When we CCA

→ Missing completly Random

→ Missing < 5%

(2000, 4)

Salary
100

CCA

(100, 4) → 20

5/4/

(900, 4) ← CCA

Result

- → Pass → 20 → 20% 10
- → Fail → 80 → 80%

Imputation

Univariate → Numerical

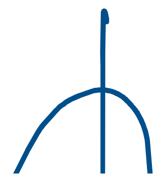
- → mean/median
- → Arbitary (Fill any value)
- → Random value

Mean/ Median

Salary
20k
30k
40k
NAN

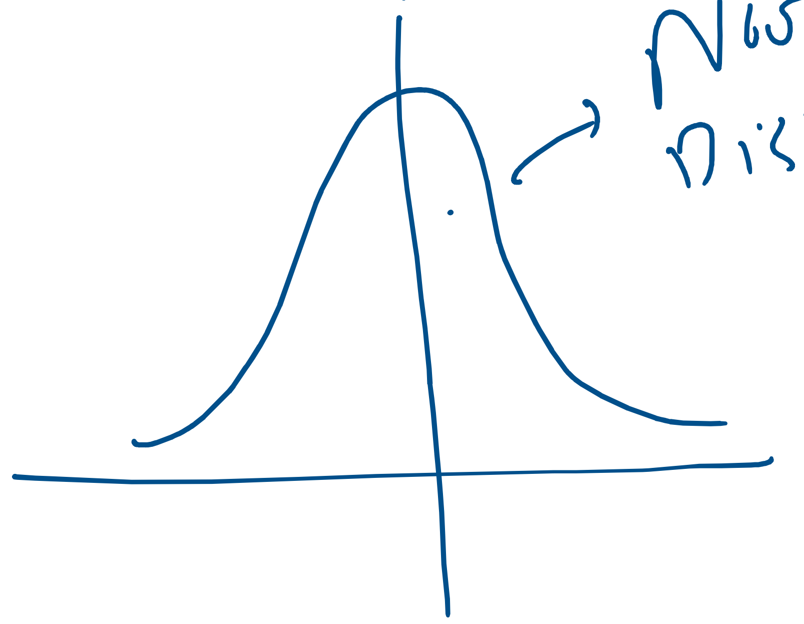
→ mean/ median

median



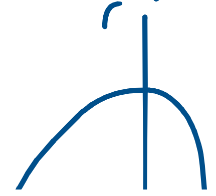
skewed

mean

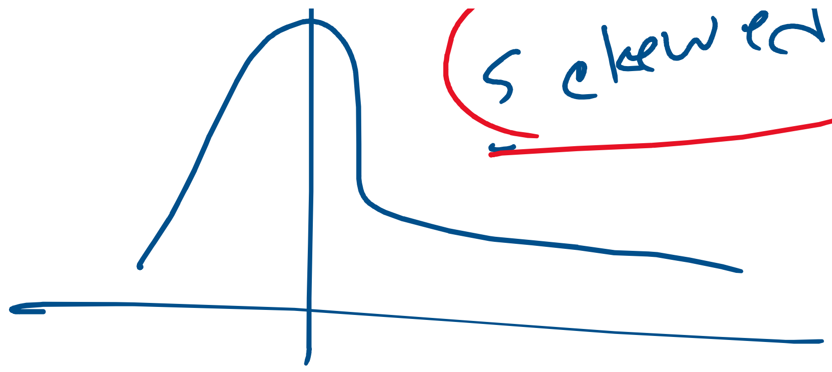


→ Normal Distributi

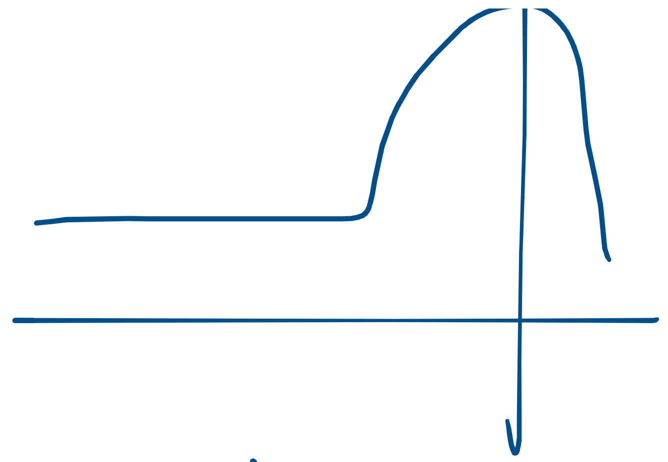
median



NAN
35 K
35 K
500 K



Skewed



Advantage

1) Simple calculation
2) M < 5%

DisAdvantage

1) Distribution shape change ✓
2) Outliers → ✓
3) Corelation change ✓

When to use

1) Missing compeletly Random
2) M < 5%

Arbitary → salary → (0, -1, 99)

Missing →

Random value

Salary

Advantage

knowledge

NAN

NAN

→ Simple

→ Distribution same

→ correlation

categorical data

→ most frequent value

→ "Missing"